There are some changes in which heat is absorbed. For example, water absorbs heat when it evaporates to form water vapours. Similarly ice absorbs heat when it melts to form water. Such changes in which heat is absorbed are known as endothermic changes. Dissolution of glucose in water is also an endothermic change.

3.8 Periodic and non-periodic change

Depending on whether or not a change repeats itself after a definite period of time, it can be classified as periodic change or a non-periodic change.

Periodic changes

Changes that repeat themselves after a definite interval of time are called **periodic changes.**

Rotation and Revolution of earth, beating of the heart, clock striking every hour, motion of the seconds-hand / minute-hand / hour-hand of a clock are some examples of periodic changes.

Every year we observe that seasons changes. We go from rains to winter and winter to summer and so on.

- What types of clothes are worn in winter?

- What are the clothes that we wear in summer? ---

If the winter season changes into summer, we observe change in the texture type of clothes we wear. We wear woolen clothes in winter and cotton clothes in summer. Similarly, we observe that the winter season is cool and summer season is hot. In winter, duration of night is longer than in summer. We take cold drinks in summer but prefer hot tea, coffee or milk in winter. These changes that we observe show the change of seasons.

The seasons and changes in weather occur because earth rotates on its fixed axis. Changing seasons are almost periodic in nature.

Non-periodic changes

Changes that do not repeat themselves after a definite interval of time and occur randomly are called **non-periodic changes**. Eruption of a volcano, occurrence of an earthquake, a streak of lighting flash across the sky during a thunderstorm, running of a batsman between the wickets, movement of legs while dancing are a few examples of non-periodic changes.

POINTS TO REMEMBER

- Particle arrangement within the state of matter gets disturbed upon heating. The disturbance is seen either as expansion or contraction.
- A process in which liquid changes into vapour on heating is called evaporation.
- A process in which solid changes into liquid on heating is called melting or fusion.
- A process in which gas changes into a liquid is called condensation.
- A process in which liquid changes into solid is called freezing.
- Physical changes are the changes in which only physical properties of a substance undergo a change and there is no change in its chemical composition.
- Solid substances are usually purified by the process of crystallization.
- Evaporation is the technique used to separate dissolved solids from a solid-liquid mixture.
- Certain solid substances like camphor, naphthalene get converted into gas directly without becoming liquid upon heating by sublimation.

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